

Lecture 3: Pivoting and $PA = LU$

Math 6610

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- See how one row exchange rescues the 10^{-20} example

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- Understand the bookkeeping $L_{n-1}P_{n-1} \cdots L_1P_1A = U$ and how it collapses to $PA = LU$
- State the partial pivoting algorithm and its guarantee $|l_{jk}| \leq 1$

Recap I

Example. Continuing from last time...

$$\tilde{L}\tilde{y} = b \quad \tilde{y} = \begin{bmatrix} 1 \\ 10^{-20} \end{bmatrix}$$

$$\tilde{U}\tilde{x} = \tilde{y} \quad \tilde{x} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \neq x$$

The culprit is 10^{20} .

Pivoting I

$$A = \begin{bmatrix} 10^{-20} & 1 \\ 1 & 1 \end{bmatrix} \quad P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad (\text{permutation matrix})$$

$$PA = \begin{bmatrix} 1 & 1 \\ 10^{-20} & 1 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 \\ 10^{-20} & 1 \end{bmatrix}}_{=L} \underbrace{\begin{bmatrix} 1 & 1 \\ 0 & 1 - 10^{-20} \end{bmatrix}}_{=U}$$

Floating point arithmetic:

$$\tilde{L} = L \quad \tilde{U} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$PAx = Pb \quad LUx = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\tilde{L}\tilde{y} = Pb \quad \tilde{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\tilde{U}\tilde{x} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \tilde{x} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \approx x = A^{-1}b$$

Pivoting II

Every time you use Gaussian elimination you need to use pivoting.

Example (General Pivoting Example). $n = 4, k = 2$ (4×4 matrix)

$$U = \begin{bmatrix} * & * & * & * \\ 0 & \boxed{*} & * & * \\ 0 & * & * & * \\ 0 & * & * & * \end{bmatrix} \xrightarrow{P_2} \begin{bmatrix} * & * & * & * \\ 0 & \# & \# & \# \\ 0 & * & * & * \\ 0 & \# & \# & \# \end{bmatrix} \quad \text{exchange rows 2 \& 4}$$
$$\xrightarrow{L_2} \begin{bmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & \# & \# \\ 0 & 0 & \# & \# \end{bmatrix}$$

In general:

$$L_{n-1}P_{n-1} \cdots L_3L_2P_2L_1P_1A = U$$

If you don't need to pivot, then use $P_i = I$ (the identity).

Pivoting III

Example (*Specific Pivoting Example*).

$$A = \begin{bmatrix} 2 & 1 & 1 & 0 \\ 4 & 3 & 3 & 1 \\ 8 & 7 & 9 & 5 \\ 6 & 7 & 9 & 8 \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{P_1} A = \begin{bmatrix} 8 & 7 & 9 & 5 \\ 4 & 3 & 3 & 1 \\ 2 & 1 & 1 & 0 \\ 6 & 7 & 9 & 8 \end{bmatrix}$$

$$L_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 & 0 \\ -\frac{1}{4} & 0 & 1 & 0 \\ -\frac{3}{4} & 0 & 0 & 1 \end{bmatrix}$$

Pivoting IV

$$L_1 P_1 A = \begin{bmatrix} 8 & 7 & 9 & 5 \\ 0 & \frac{7}{4} & \frac{9}{4} & \frac{17}{4} \\ 0 & -\frac{3}{4} & -\frac{5}{4} & -\frac{5}{4} \\ 0 & -\frac{1}{2} & -\frac{3}{2} & -\frac{3}{2} \end{bmatrix}$$

$\vdots$

$$U = \begin{bmatrix} 8 & 7 & 9 & 5 \\ 0 & \frac{7}{4} & \frac{9}{4} & \frac{17}{4} \\ 0 & 0 & -\frac{6}{7} & -\frac{2}{7} \\ 0 & 0 & 0 & \frac{2}{3} \end{bmatrix}$$

Example (Pivoting in Matrix Form). $n = 4$:

$$L_3 P_3 \underbrace{I}_{=P_3^T P_3} L_2 P_2 L_1 \underbrace{I}_{=P_2^T P_3^T P_3 P_2} P_1 A = U$$

$$\underbrace{L_3}_{=L'_3} \underbrace{(P_3 L_2 P_3^T)}_{=L'_2} \underbrace{(P_3 P_2 L_1 P_2^T P_3^T)}_{=L'_1} \underbrace{(P_3 P_2 P_1)}_{=P} A = U$$

Pivoting V

$$L_3' L_2' L_1' (PA) = U$$

$L_1' (\neq L_1^T)$ has the same structure as L_1 , with nontrivial elements in column 1 reordered.

$$L_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 & 0 \\ -\frac{1}{4} & 0 & 1 & 0 \\ -\frac{3}{4} & 0 & 0 & 1 \end{bmatrix}, \quad L_1' = \begin{bmatrix} 1 & 0 & 0 & 0 \\ * & 1 & 0 & 0 \\ * & 0 & 1 & 0 \\ * & 0 & 0 & 1 \end{bmatrix}$$

$$PA = \underbrace{(L_1')^{-1} (L_2')^{-1} (L_3')^{-1}}_{=L} U$$

General case:

$$PA = LU$$

where P is the permutation matrix and L and U are lower and upper triangular matrices, respectively.

Remark (Algorithm: LU factorization with partial pivoting). Input: $A \in \mathbb{C}^{n \times n}$
Set $U = A$, $L = I$, $P = I$.

Pivoting VI

- for $k = 1, 2, \dots, n - 1$
 - ▶ choose $i \geq k$ such that $|u_{ik}| = \max_{k \leq l \leq n} |u_{lk}|$
 - ▶ if $u_{kk} = 0$, stop: A is singular
 - ▶ $u_{k,k:n} \leftrightarrow u_{i,k:n}$ (interchange rows i and k)
 - ▶ $l_{k,1:k-1} \leftrightarrow l_{i,1:k-1}$
 - ▶ $p_{k,:} \leftrightarrow p_{i,:}$
 - ▶ for $j = k + 1, k + 2, \dots, n$
 - * $l_{jk} = u_{jk} / u_{kk}$
 - * $u_{j,k:n} = u_{j,k:n} - l_{jk} u_{k,k:n}$
 - ▶ end (j)
- end (k)

Output: U, L, P such that

$$PA = LU$$

where $L = [l_{jk}]$ with $|l_{jk}| \leq 1$ for all $j > k$.

Some Remarks: Why $L'_i = PL_iP^T$ Keeps Its Shape I

The reordering trick inserted $I = P^TP$ between factors. The conjugated matrices are, e.g.,

$$L'_1 = (P_3P_2) L_1 (P_3P_2)^T.$$

Why does this stay a Frobenius matrix for column 1? Write $L_1 = I + c e_1^T$ with c supported on entries $2, \dots, n$, and let $\Pi := P_3P_2$. Both P_2 and P_3 only permute rows/columns *from index 2 onward* (step k pivots among rows $\geq k$), so $\Pi e_1 = e_1$. Then

$$\Pi L_1 \Pi^T = \Pi(I + c e_1^T) \Pi^T = I + (\Pi c) \underbrace{(\Pi e_1)^T}_{=e_1^T} = I + (\Pi c) e_1^T.$$

So L'_1 is again unit lower-triangular with its column-1 multipliers merely *reordered* ($c \mapsto \Pi c$) – exactly the structure claimed on the previous slide.

Some Remarks: Why $L'_i = PL_iP^T$ Keeps Its Shape II

Permutation matrix facts used here:

- P has exactly one 1 in each row and column; $P^T P = I$, so $P^{-1} = P^T$.
- Left multiplication PA permutes *rows*; right multiplication AP^T permutes *columns* the same way.
- Products of permutation matrices are permutation matrices, so $P := P_{n-1} \cdots P_2 P_1$ in $PA = LU$ is one single row reordering of A , decided during elimination.

Cost of pivoting: step k scans $n - k + 1$ candidate pivots, so the total is $O(n^2)$ comparisons – negligible next to $\frac{2n^3}{3}$ flops. Stability is (essentially) free.

Why $|l_{jk}| \leq 1$, and Why We Care

At step k we choose the pivot row $i \geq k$ with the *largest* $|u_{ik}|$, then form

$$l_{jk} = \frac{u_{jk}}{u_{kk}} \quad (j > k).$$

After the swap, $|u_{kk}| = \max_{l \geq k} |u_{lk}| \geq |u_{jk}|$, hence $|l_{jk}| \leq 1$ automatically.

This bounds $\|L\|$ by a modest constant and prevents the 10^{20} -type blow-up in L . (Whether U can still grow is the “growth factor” question – we settle it in Lecture 15.)

Practical note: in MATLAB, $[L,U,P] = \text{lu}(A)$ returns exactly the objects of today’s algorithm, and $A \setminus b$ performs $PA = LU$ plus two triangular solves under the hood.

Looking Ahead

- **Next lecture:** Newton's Method: Derivation